

DS ASSIGNMENT SET 3

-Cristina Jane B

1. Write a C program to store n elements in an array using dynamic memory allocation and find the largest number.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main()
```

```
{
```

```
    int n, i, largest;
```

```
    int *arr;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    arr = (int *)malloc(n * sizeof(int));
```

```
    printf("Enter %d elements:\n", n);
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    largest = arr[0];
```

```
    for(i = 1; i < n; i++)
```

```
    {
```

```

        if(arr[i] > largest)
        {
            largest = arr[i];
        }
    }

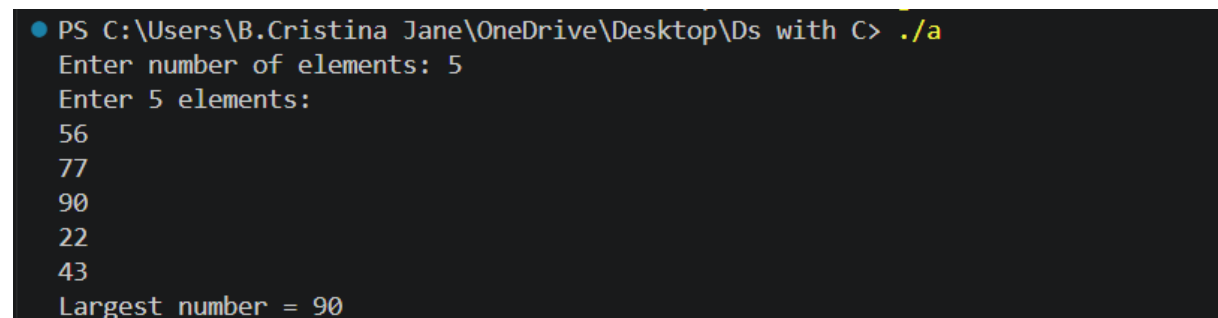
    printf("Largest number = %d", largest);

    free(arr);

    return 0;
}

```

OUTPUT:



```

PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds with C> ./a
Enter number of elements: 5
Enter 5 elements:
56
77
90
22
43
Largest number = 90

```

2.print an array in reverse order using pointers

```
#include <stdio.h>
```

```

int main()
{
    int arr[5], i;
    int *p;

    p = arr;

```

```

printf("Enter 5 elements:\n");

for(i = 0; i < 5; i++)
{
    scanf("%d", p + i);
}

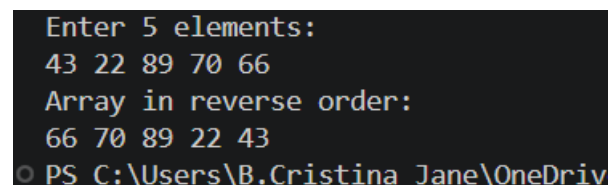
printf("Array in reverse order:\n");

for(i = 4; i >= 0; i--)
{
    printf("%d ", *(p + i));
}

return 0;
}

```

OUTPUT:



```

Enter 5 elements:
43 22 89 70 66
Array in reverse order:
66 70 89 22 43
PS C:\Users\B.Cristina Jane\OneDrive

```

3. Write a C program to store information of students using structure and sort them by Name.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
struct Student
```

```
{
```

```
char name[50];

int rollno;

float mark;

};

int main()
{
    struct Student s[5], temp;

    int i, j;

    printf("Enter details of 5 students:\n");

    for(i = 0; i < 5; i++)
    {
        printf("\nStudent %d\n", i + 1);

        printf("Enter name: ");
        scanf("%s", s[i].name);

        printf("Enter roll number: ");
        scanf("%d", &s[i].rollno);

        printf("Enter mark: ");
        scanf("%f", &s[i].mark);
    }

    /* Sorting by name */
    for(i = 0; i < 5; i++)
    {
        for(j = i + 1; j < 5; j++)
```

```

    {
        if(strcmp(s[i].name, s[j].name) > 0)
        {
            temp = s[i];
            s[i] = s[j];
            s[j] = temp;
        }
    }
}

printf("\nStudents sorted by Name:\n");

for(i = 0; i < 5; i++)
{
    printf("\nName    : %s", s[i].name);
    printf("\nRoll No  : %d", s[i].rollno);
    printf("\nMark     : %.2f\n", s[i].mark);
}

return 0;
}

```

OUTPUT:

PS C:\Users\B.Cristina Jane\OneDrive\Desktop>

Enter details of 5 students:

Student 1

Enter name: kala

Enter roll number: 101

Enter mark: 100

Student 2

Enter name: rama

Enter roll number: 102

Enter mark: 99

Student 3

Enter name: jane

Enter roll number: 103

Enter mark: 100

Student 4

Enter name: jeni

Enter roll number: 104

Enter mark: 67

Student 5

Enter name: ravi

Enter roll number: 105

Enter mark: 56

Students sorted by Name:

Name : jane
Roll No : 103
Mark : 100.00

Name : jeni
Roll No : 104
Mark : 67.00

Name : kala
Roll No : 101
Mark : 100.00

Name : rama
Roll No : 102
Mark : 99.00

Name : ravi
Roll No : 105
Mark : 56.00

4. Write a C program to add 'n' distances given in inch- feet system using structures

```
#include <stdio.h>
```

```
struct Distance
```

```
{  
    int feet;  
    int inch;  
};
```

```
int main()
```

```
{  
    int n, i;  
    struct Distance d[100], sum = {0, 0};  
  
    printf("Enter number of distances: ");  
    scanf("%d", &n);  
  
    printf("Enter the distances:\n");  
  
    for(i = 0; i < n; i++)  
    {  
        printf("Distance %d - Feet: ", i + 1);  
        scanf("%d", &d[i].feet);  
  
        printf("Distance %d - Inches: ", i + 1);  
        scanf("%d", &d[i].inch);  
  
        sum.feet = sum.feet + d[i].feet;  
        sum.inch = sum.inch + d[i].inch;  
    }  
}
```

```

sum.feet = sum.feet + sum.inch / 12;

sum.inch = sum.inch % 12;

printf("\nTotal Distance = %d feet %d inches",
       sum.feet, sum.inch);

return 0;
}

```

OUTPUT:

```

Enter number of distances: 2
Enter the distances:
Distance 1 - Feet: 10
Distance 1 - Inches: 55
Distance 2 - Feet: 20
Distance 2 - Inches: 33

Total Distance = 37 feet 4 inches
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\Ds with C>

```

5. Write a C program to add two complex numbers by passing structure to a function.

```
#include <stdio.h>
```

```
struct Complex
```

```

{
    float real;
    float imag;
};

```

```
struct Complex add(struct Complex c1, struct Complex c2)
```

```

{
    struct Complex result;

```



```
    result.real = c1.real + c2.real;

    result.imag = c1.imag + c2.imag;


    return result;
}


int main()
{
    struct Complex c1, c2, sum;


    printf("Enter first complex number:\n");
    printf("Real part: ");
    scanf("%f", &c1.real);


    printf("Imaginary part: ");
    scanf("%f", &c1.imag);


    printf("\nEnter second complex number:\n");
    printf("Real part: ");
    scanf("%f", &c2.real);


    printf("Imaginary part: ");
    scanf("%f", &c2.imag);


    sum = add(c1, c2);


    printf("\nSum = %.2f + %.2fi",
        sum.real, sum.imag);


    return 0;
```

```
}
```

OUTPUT:

```
PS C:\Users\B.C.Rishina> gcc oneDrive\desktop\
Enter first complex number:
Real part: 5
Imaginary part: 8

Enter second complex number:
Real part: 5
Imaginary part: 9

Sum = 10.00 + 17.00i
```

6. Write a C program to maintain an inventory management system using structure.

```
#include <stdio.h>
```

```
struct Product
```

```
{
```

```
    int id;
```

```
    char name[50];
```

```
    int quantity;
```

```
    float price;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Product p[100];
```

```
    int n, i;
```

```
    printf("Enter number of products: ");
```

```
    scanf("%d", &n);
```

```
    for(i = 0; i < n; i++)
```

```
{  
  
    printf("\nEnter details of Product %d\n", i + 1);  
  
    printf("Product ID: ");  
    scanf("%d", &p[i].id);  
  
    printf("Product Name: ");  
    scanf("%s", p[i].name);  
  
    printf("Quantity: ");  
    scanf("%d", &p[i].quantity);  
  
    printf("Price: ");  
    scanf("%f", &p[i].price);  
}
```

```
printf("\n----- INVENTORY DETAILS ----- \n");
```

```
printf("ID\tName\tQuantity\tPrice\n");
```

```
for(i = 0; i < n; i++)  
{  
    printf("%d\t%s\t%d\t\t%.2f\n",  
        p[i].id,  
        p[i].name,  
        p[i].quantity,  
        p[i].price);  
}
```

```
return 0;
```

```
}
```

OUTPUT:

```
Enter number of products: 2

Enter details of Product 1
Product ID: 101
Product Name: tv
Quantity: 3
Price: 5000

Enter details of Product 2
Product ID: 102
Product Name: phone
Quantity: 5
Price: 2000

----- INVENTORY DETAILS -----
ID      Name    Quantity  Price
101     tv       3         5000.00
102     phone    5         2000.00
```

7. Write a C program for decimal to binary, octal and hexadecimal conversion.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter a decimal number: ");
```

```
    scanf("%d", &n);
```

```
    printf("Binary    : ");
```

```
    int temp = n;
```

```
    int binary[32], i = 0;
```

```

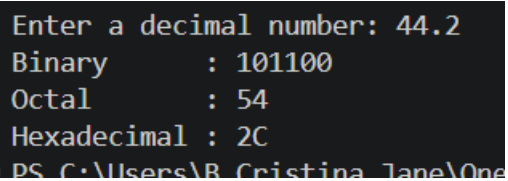
if(temp == 0)
{
    printf("0");
}
else
{
    while(temp > 0)
    {
        binary[i] = temp % 2;
        temp = temp / 2;
        i++;
    }

    for(i = i - 1; i >= 0; i--)
    {
        printf("%d", binary[i]);
    }
}

printf("\nOctal    : %o", n);
printf("\nHexadecimal : %X", n);

return 0;
}

```



```

Enter a decimal number: 44.2
Binary      : 101100
Octal       : 54
Hexadecimal : 2C
PS C:\Users\R. Cristina Lane\One

```

8. Write a C program to calculate the difference between two time periods.

```
#include <stdio.h>
```

```
struct Time
```

```
{
```

```
    int hour;
```

```
    int minute;
```

```
    int second;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Time t1, t2, diff;
```

```
    printf("Enter first time:\n");
```

```
    printf("Hour: ");
```

```
    scanf("%d", &t1.hour);
```

```
    printf("Minute: ");
```

```
    scanf("%d", &t1.minute);
```

```
    printf("Second: ");
```

```
    scanf("%d", &t1.second);
```

```
    printf("\nEnter second time:\n");
```

```
    printf("Hour: ");
```

```
    scanf("%d", &t2.hour);
```

```
    printf("Minute: ");
```

```
    scanf("%d", &t2.minute);
```

```
printf("Second: ");
scanf("%d", &t2.second);

if(t2.second < t1.second)
{
    t2.second += 60;
    t2.minute--;
}

if(t2.minute < t1.minute)
{
    t2.minute += 60;
    t2.hour--;
}

diff.second = t2.second - t1.second;
diff.minute = t2.minute - t1.minute;
diff.hour = t2.hour - t1.hour;

printf("\nTime Difference = %d:%d:%d",
       diff.hour, diff.minute, diff.second);

return 0;
}
```

OUTPUT:

```
PS C:\Users\B.Cristina Jane\OneDrive\Desktop\DS w...
Enter first time:
Hour: 2
Minute: 44
Second: 30

Enter second time:
Hour: 4
Minute: 90
Second: 56

Time Difference = 2:46:26
```

9. Write a C program to implement an employee management system using structures.

```
#include <stdio.h>
```

```
struct Employee
```

```
{
```

```
    int id;
```

```
    char name[50];
```

```
    float salary;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Employee e[100];
```

```
    int n, i;
```

```
    printf("Enter number of employees: ");
```

```
    scanf("%d", &n);
```

```
    for(i = 0; i < n; i++)
```

```
    {
```



```

printf("\nEnter details of Employee %d\n", i + 1);

printf("Employee ID: ");
scanf("%d", &e[i].id);

printf("Employee Name: ");
scanf("%s", e[i].name);

printf("Salary: ");
scanf("%f", &e[i].salary);
}

printf("\n----- EMPLOYEE DETAILS ----- \n");
printf("ID\tName\tSalary\n");

for(i = 0; i < n; i++)
{
    printf("%d\t%s\t%.2f\n",
        e[i].id,
        e[i].name,
        e[i].salary);
}

return 0;
}

```

OUTPUT:

```
Enter number of employees: 2
```

```
Enter details of Employee 1
```

```
Employee ID: 101
```

```
Employee Name: saru
```

```
Salary: 5000
```

```
Enter details of Employee 2
```

```
Employee ID: 102
```

```
Employee Name: ella
```

```
Salary: 4000
```

```
----- EMPLOYEE DETAILS -----  
ID      Name      Salary  
101     saru       5000.00  
102     ella       4000.00
```

10. Write a C program to calculate the area and perimeter of a rectangle and triangle using function pointers.

```
#include <stdio.h>
```

```
float rectangleArea(float l, float b)
```

```
{  
    return l * b;  
}
```

```
float rectanglePerimeter(float l, float b)
```

```
{  
    return 2 * (l + b);  
}
```

```
float triangleArea(float b, float h)
```

```
{  
    return 0.5 * b * h;  
}
```

```
float trianglePerimeter(float a, float b, float c)
```

```
{
```

```
    return a + b + c;
```

```
}
```

```
int main()
```

```
{
```

```
    float l, b, h;
```

```
    float a, c;
```

```
    float (*area)(float, float);
```

```
    float (*perimeter)(float, float);
```

```
    printf("Enter length and breadth of rectangle: ");
```

```
    scanf("%f %f", &l, &b);
```

```
    area = rectangleArea;
```

```
    perimeter = rectanglePerimeter;
```

```
    printf("\nRectangle Area = %.2f", area(l, b));
```

```
    printf("\nRectangle Perimeter = %.2f", perimeter(l, b));
```

```
    printf("\n\nEnter base and height of triangle: ");
```

```
    scanf("%f %f", &b, &h);
```

```
    area = triangleArea;
```

```
    printf("\nTriangle Area = %.2f", area(b, h));
```

```
    printf("\n\nEnter three sides of triangle: ");
```

```
scanf("%f %f %f", &a, &b, &c);
```

```
printf("Triangle Perimeter = %.2f",
```

```
trianglePerimeter(a, b, c));
```

```
return 0;
```

```
}
```

OUTPUT:

```
Enter length and breadth of rectangle: 12 10
```

```
Rectangle Area = 120.00
```

```
Rectangle Perimeter = 44.00
```

```
Enter base and height of triangle: 20 10
```

```
Triangle Area = 100.00
```

```
Enter three sides of triangle: 10 10 10
```

```
Triangle Perimeter = 30.00
```